

Towards the Development of a Wearable Device Enhancing Inclusivity and Monitoring for Dementia Patients

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Abstract. The rapidly aging global population presents significant challenges, particularly in managing neurodegenerative diseases such as dementia. This paper discusses the development of a comprehensive wearable device system designed to monitor and care for elderly individuals with dementia. This study focuses on the identification of relevant functional requirements and presents an initial prototype of the envisioned device. By integrating advanced sensor technologies and artificial intelligence, the system aims to ensure accurate and continuous monitoring of health indicators and daily activities. Its design approach emphasizes usability and non-intrusiveness, making it suitable for continuous wear by elderly users. The device incorporates sensors to collect data on heart rate, blood oxygen levels, and daily activities. These data are processed using a variety of machine learning techniques to provide reliable activity recognition and health monitoring, aiming to enhance the quality of life and facilitate timely medical interventions for both patients and caregivers. Preliminary results demonstrate the system's potential in providing dependable health monitoring and activity detection, which could significantly aid in managing dementia.

Keywords: Wearable Devices, Dementia, Artificial Intelligence, Inclusive Design

1 Introduction

The increase in the number of elderly people is a growing global phenomenon. The World Health Organization projects that by 2030, one in six people will be aged 60 or over, and this ratio could double by 2050 [1]. This demographic growth can be attributed above all to the increase in average life expectancy and constant improvements in living conditions. However, since ageing is the main factor in the development of

neurodegenerative diseases, the prevalence of these has also increased significantly. Dementia, which arise due to the progressive degeneration of the cognitive system, represent a high-risk condition for people who reach older ages[2] According to the Diagnostic and Statistical Manual of Mental Disorders (DSM-5), “Major Neurocognitive Disorder”, better known as dementia, is a clinical syndrome characterized by progressive cognitive decline affecting various domains such as memory, functional execution, language, learning, perceptual-motor or social cognition [3]

Dependence on formal and informal caregivers manifests itself early in these patients due to the gradual deterioration of functional capacities, compromising the execution of basic daily activities [4] [5]. Technology has gained a potential role in diagnosing and monitoring diseases in a safer and more controlled way [6]. In the case of dementia, the intervention of continuous monitoring systems has aided in monitoring daily activities, helping caregivers to provide more active and appropriate care as dementia progresses [6], [7]

In a generic description, this study is a part of a large project which aims to develop and validate a new system for location and activity monitoring, promoting active, healthy, and inclusive ageing combining technology and design as approaches to a product suited to the needs of people with dementia and their caregivers. This new system will feature two wearable data acquisition devices, the HowMi [8] (in-ear) and a wristband represented in **Fig. 1**. These devices will acquire biometric data such as body temperature, heart rate, blood oxygen saturation level, as well as indoor and outdoor location. The data will be sent to a server, where it will be managed and monitored. The information will be made available on a web portal for institutional professionals, caregivers, and family members, helping them to decide on the most appropriate type of care for the elderly person’s situation.

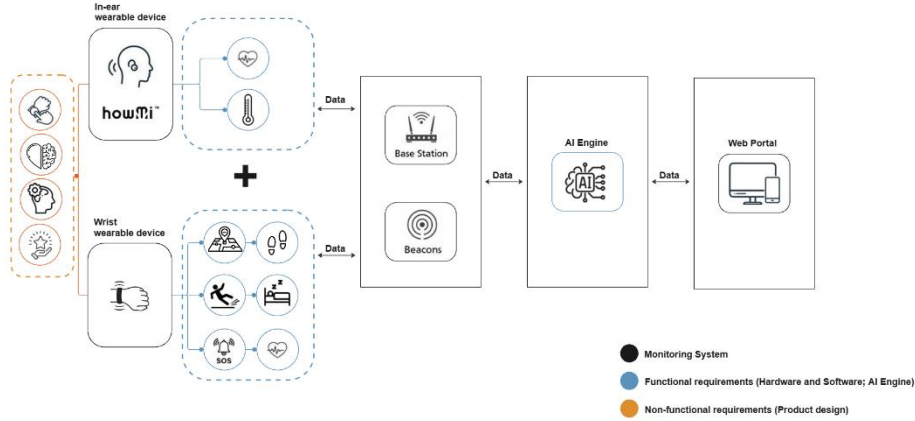


Fig. 1. Platform for routine measurement and monitoring overview

This paper is organized in 3 sections, Section 2, “Methodology”, describes the approaches and methods used to achieve the stipulated objectives of the current development of the work, Section 3, ”Development”, describes the result of all the progress

made in the development of the devices in terms of hardware and software, artificial intelligence and design based on the methodologies outlined, and Section 4 “Final Remarks” describes the next steps that will be taken to continue the work according to the methodologies defined and the results achieved.

2 Methodology

This work used an iterative, elderly user-centric approach to develop a wearable monitoring device, with a focus on safety, efficiency and usability. The process included the integrated development of hardware and software, in addition to the physical design of the wearable.

2.1 Hardware

The efficacy of the AI models depends significantly on the quality and precision of the data collected by the hardware. The wrist wearable device utilizes several sensors to obtain biological, location and movement information, such as heart rate (HR), blood oxygen saturation (SPO2), outdoor location, acceleration and angular velocity. These non-invasive sensors aim to provide an accurate estimate of the patient's condition and evolution. The device architecture is illustrated in **Fig. 2**. Subsequently, the various components that constitute the hardware will be described in detail.

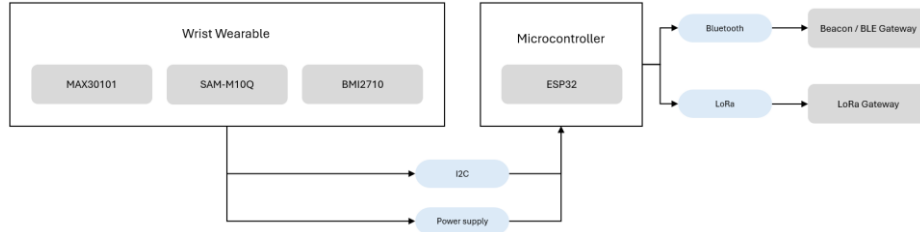


Fig. 2. Wrist wearable device architecture

The MAX30101 [9] sensor implements photoplethysmography (PPG) to measure heart rate and peripheral oxygen saturation (SpO2). The sensor features integrated LEDs (red, green, infrared), photodetectors, optical components as well as ambient light cancellation in a compact design (5.6mm × 3.3mm × 1.55mm), optimized for ultra-low-power operations suitable for mobile health monitoring devices.

The SAM-M10Q [10] Global Navigation Satellite System (GNSS) module provides precise location data. It includes a built-in antenna, operates with low power consumption, and is designed for integration into compact devices (15.5mm × 15.5mm × 6.3mm), enhancing reliability in signal-obstructed environments.

The selected Inertial Measurement Unit (IMU), the BMI270 [11], is a high-performance, ultra-low-power device that integrates a 3-axis accelerometer and a 3-axis gyroscope. Additionally, it supports multiple communication interfaces, such as I2C and

SPI, ensuring seamless integration into various systems. Its compact size (2.5mm × 3.0mm × 0.8mm) and efficiency makes it particularly suitable for wearable and consumer electronics.

Data processing and communications via Bluetooth and LoRa (Long Range) are handled by the Heltec Wifi LoRa 32 [12] development board equipped with an ESP32-S3FN8 [13] microcontroller. This board features integrated connectivity options, measuring 44mm × 30.5mm × 6.8mm, and supporting portable and mobile health applications.

2.2 Product Design

Regarding the monitoring system, functional and non-functional requirements were identified as the basis for developing a wearable device suitable for users represented in **Table 1**. The functional requirements focus on the functional part that the device can offer, containing as many features as possible that are ideal for monitoring elderly people with dementia. The non-functional requirements focus on aspects that the device can contain on a visual, physical, emotional or other level, facilitating interaction with the user through an empathetic approach [7]. Thus, it is not only highly important to understand that the wrist is a minimally intrusive measurement site, but also a strategic point for monitoring vital signs, as it is close to the blood vessels and movements are more controlled. Therefore, all the data to be measured had to be studied so that the layout of the sensors was properly organized. Heart rate is data that can be measured on certain parts of the body and for the sensor to be able to capture the most accurate data, it must be in contact with the skin.

Table 1. Functional and non-functional requirements of wearable devices for elderly with dementia.

Typology	Category	Requirement
Functional Requirements	Hardware / Software	Real-time location
		Steps counting
		Fall detection
		Sleep monitoring
		SOS situation
		Heart rate measurement
Non-Functional Requirements	Physical Domain	Oxygen saturation
		Comfort
		Durability
		Tangibility
	Emotional Domain	Visibility
		Aesthetics
		Equitable use
		Autonomy
		Customization

Cognitive Domain	Usability
Implementation	Reliability
Success Domain	Context
	Price

In order to define the shape of the device, an analysis was carried out of possible shapes that the device could contain according to what exists commercially. This analysis brought together positive and negative points, and it is important to highlight that because the device is aimed at elderly with dementia, the acceptance of its integration must also be done through physical elements [14] In addition, it was important to study the geometry of the wrist so that the sensors could be correctly positioned, guaranteeing safe data collection without interference. However, even though all wrists have different anthropometry, the device should be suitable for the maximum number of users.

This study led to the creation of three-dimensional models using the 3D modeling software, SolidWorks [15]. These models were then tested using rapid prototyping methods, namely 3D Fused Deposition Modeling printing.

2.3 Data Acquisition

Data acquisition is a critical phase where the wearable's sensors are rigorously tested to validate their accuracy and reliability in capturing physiological and movement data.

Reflective pulse oximetry uses LEDs to illuminate the skin tissue and a photodiode to detect the reflected signal. This signal contains light modulated by volumetric changes in the arteries and capillaries within the tissue. PPG signals consist of a DC (Direct Current) component and an AC (Alternating Current) component. The non-pulsatile tissue absorbs light and generates the DC component, while the AC component is attributed to the pulsatile nature of arterial blood.

The instantaneous heart rate is determined by measuring the time interval between consecutive systolic peaks, derived solely from the green signal. The following steps, in **Fig. 3**, outline the basic calculation of instantaneous heart rate.

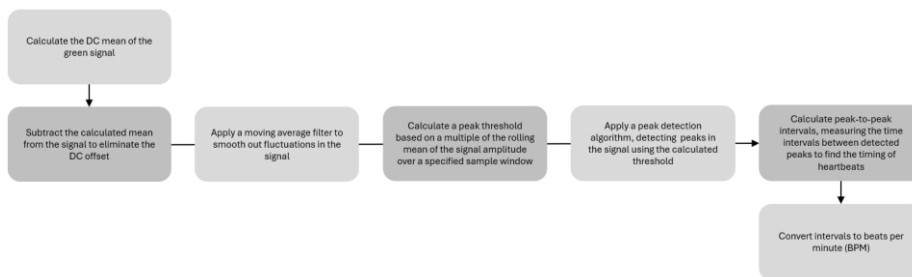


Fig. 3. Flowchart of heart rate calculation process

The SpO2 value is computed using both RED and IR waveforms, following the algorithm provided by the manufacturer [16]. First, the DC and AC components are

extracted from both signals. Then, the normalized ratio (R) of the RED waveform to the IR waveform is calculated. This ratio (R), as described by equation (1), is possible because the proportion of oxygenated hemoglobin in the blood affects the absorbance ratio of infrared to red light within the PPG sensor.

$$R = \frac{AC_{red}/DC_{red}}{AC_{ir}/DC_{ir}} \quad (1)$$

Where, AC_{red} represents the Alternating Current (AC) component of the red signal, DC_{red} represents the Direct Current (DC) component of the red signal, AC_{ir} represents the AC component of the infrared signal, and DC_{ir} represents the DC component of the infrared signal.

The ratio (R) derived from these components can be used to empirically determine the SpO2 value through a lookup table or a curvilinear approximation, as provided by the manufacturer [16]. The SpO2 calculation follows the manufacturer's calibration [16], described in equation (2).

$$SpO2 = 1.5958422 \times R^2 - 34.6596622 \times R + 112.6898759 \quad (2)$$

The basic steps to calculate SpO2 are implemented in the library provided by the manufacturer as presented in **Fig. 4**.



Fig. 4. Flowchart of SpO2 calculation process

The positioning data from the SAM-M10Q can be acquired (**Fig. 5**) directly from the module's memory registers as it processes GNSS (Global Navigation Satellite System) signals internally, providing ready-to-use, high-accuracy location information. Minimal post-processing is required for standard applications.

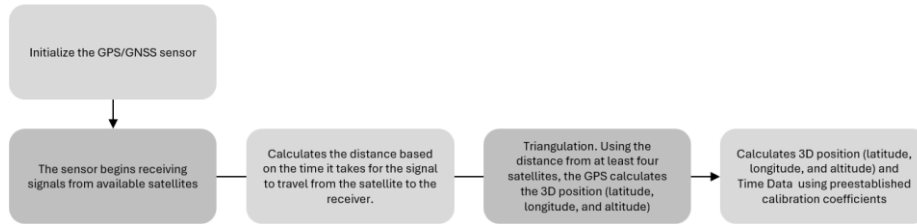


Fig. 5. Flowchart of GPS data acquisition

The motion and orientation data from the BMI270 can be acquired (**Fig. 6**) directly from the sensor's memory registers, as it is pre-calibrated for high-accuracy

measurements in dynamic environments. Only minimal post-processing is needed to convert raw data into meaningful metrics.



Fig. 6. Flowchart of inertial data acquisition

2.4 Communication Protocol

To simplify wireless integration of the device, two communication protocols were chosen, Bluetooth Low Energy (BLE) for indoor communication and LoRa for outdoor communication.

BLE was selected for indoor communication due to its low power consumption and ability to transmit data over short distances, making it ideal for communication between wearables and nearby devices.

LoRa (Long Range) was adopted for outdoor communication due to its ability to provide long-range communication with low power consumption. It is suitable for large areas, such as urban environments or outdoor settings, allowing data transmission over long distances without compromising battery life. This protocol is ideal for sending health monitoring and location data to the network in areas where BLE coverage is not available, ensuring real-time connectivity across a vast geographical area.

3 Development of an initial proof of concept

This section details the ongoing enhancements to the wearable device, designed to monitor elderly individuals with dementia. We focus on refining the device's hardware and software to improve functionality and user-friendliness, while integrating advanced sensor technology and artificial intelligence. Key developments include ergonomic re-designs for ease of use, expanded data collection for robust AI model training, and advancements in communication protocols for secure and efficient data transmission. We will also discuss the optimization of power management and data acquisition algorithms to enhance the device's operational efficiency and user safety.

3.1 Wearable Device Hardware

With the selection and study of the hardware for implementation, as well as the development flow of the wearable, a functional prototype was created using development boards of the chosen sensors divided into three parts through the middle cover represented in **Fig. 7**.

With the prototype assembled and the connections tested, the sensors were programmed individually to test the acquisition and calibration methods for the data obtained. Regarding the inertial sensor (BMI 270), it was configured to record steps, distance covered, activity and inertial data (acceleration and rotation). The GPS sensor

(SAM-M10Q) was configured to collect latitude, longitude, altitude in addition to the date and time. To monitor the heart rate in the wrist with MAX30101, the green LED was activated because the light penetrates the skin superficially, reaching the capillaries close to the surface, being well absorbed by hemoglobin, which allows the variations caused by the pulse to be detected. Furthermore, it is less sensitive to movements, reducing noise in measurements. This feature, combined with filtering algorithms, increases accuracy of measured heart rate in wearables. To measure SpO2, signals from the red and IR LEDs were processed. The value is estimated based on the relationship of red and IR signals using pre-defined calibration.

During the development, sequential data acquisition (sensor by sensor) was also tested through the application of a real-time operating system (RTOS - Real-Time Operating System). It is designed to manage tasks simultaneously, providing features that allow precise control of execution time and prioritization of tasks in systems with limited resources. To test and implement the BLE protocol, advertising of the data obtained was tested, sending data from the wearable to other devices (smartphone).

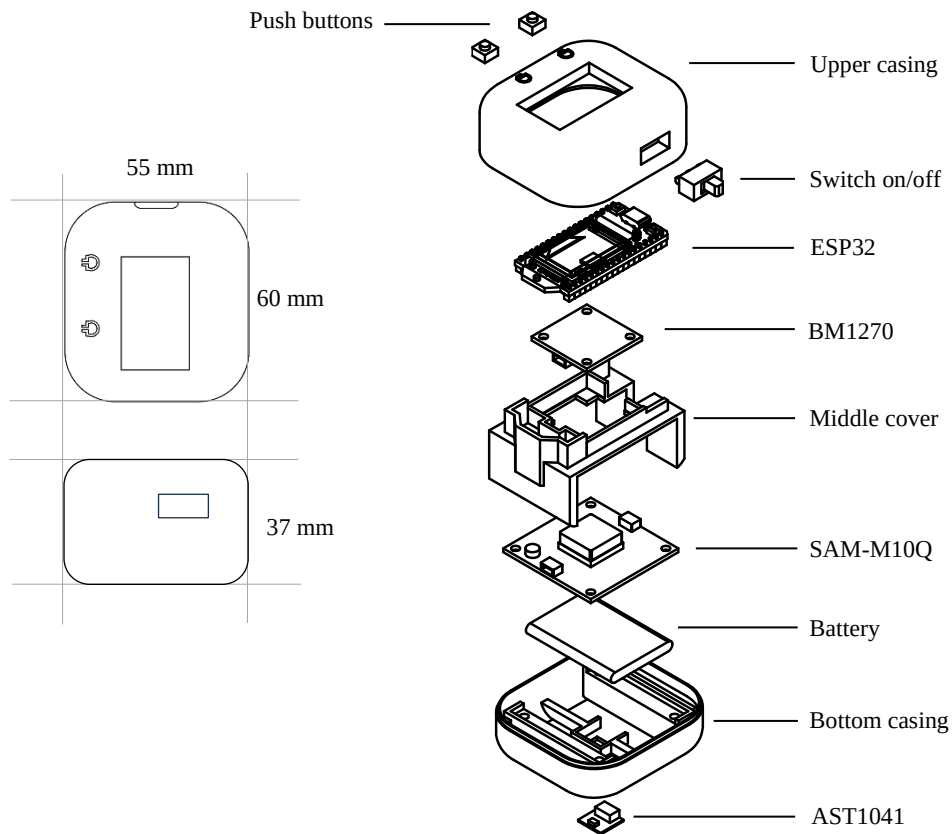


Fig. 7. System device overview and dimensions.

3.2 Wearable Device Usability and Ergonomic Design

The studies carried out on the layout and size of the sensors resulted in the creation of several virtual three-dimensional models and test prototypes. These tests made it possible to understand the constraints to be improved and to study ways of fixing each component, ensuring the greatest possible use of space. Some constraints, such as the tolerance between each component for accurate reading without interference, and the study of the size of the device according to the space occupied by the connectors between each component, were tackled continuously in order to achieve a satisfactory model. Overall, thirty prototype tests were carried out to arrive at the functional model shown in **Fig. 7**.

Regarding the study of the choice of shape and size of the display suitable for users in relation to their needs, **Table 2** and **Table 3** were drawn up addressing the advantages and disadvantages, so that the study could be demonstrated coherently. However, only the shape that the device would take was considered for this stage of the work, through the analysis carried out. In addition, for a more complete evaluation, both features will be tested with users, collecting their feedback for a more accurate choice.

3.1.1. Shape

Table 2 shows that the square/rectangular shape has more advantages for user adaptation due to its conventional shape. In addition, the compact format allows its internal geometry to be more practical, and the application of action buttons becomes more intuitive and easier to follow the design of the model. On the other hand, square/rectangular geometry contains four vertices, which, even if they can be worked with, can be uncomfortable in daily use.

In view of this, it is necessary to take into account designing a well-studied and validated format that does not generate pain during its use in users' daily lives.

Table 2. Advantages and disadvantages of existing shapes of wrist devices.

Form	Advantages	Disadvantages
Square/Rectangular	Compact form	Uncomfortable extremities
	Conventional shape	
	Versatility of button placement	
Circular	Unisex design	Button location Button design curvature
	Compact form	
	Conventional shape	
Oval/Cylindrical	Unisex design	Futuristic design Difficult adaption Button location Button design curvature Short display size
	Compact form	

3.1.2. Display size

Table 3 shows two commercially available rectangular display sizes that are suitable for adult wrist anthropometry. In this case, the fact that the device is only for viewing and not for touch interaction, contributed to the choice of the smaller display size, with the possibility of adjusting the readability of the information to be contained.

In addition, the smaller size provided a better balance between the size ratio of the wrist and the device, so that its casing would not subsequently contain even more volume and would not be uncomfortable.

Table 3. Advantages and disadvantages of existing display sizes for wearable devices.

Display size	Advantages	Disadvantages
35mmx41mm	Proportional size to the wrist Appropriate size for reading information	Disproportional size to the wrist for female users
38mmx45mm	Appropriate size for reading information	

4 Final Remarks

The work next step involves analyzing the device's current model to align its technology and design, considering wrist anthropometry to create a final model suitable for the elderly. Key elements like buttons and straps will be studied for usability, with textures and colors differentiating button functions, and a safety system added to the strap to prevent accidental removal.

Development will also focus on creating an extensive dataset for monitoring dementia, integrating advanced AI techniques to improve real-time monitoring. Data will be structured and stored via BLE on smartphones.

Efforts will prioritize optimizing data acquisition algorithms, reducing power consumption, and enhancing communication protocols, including testing LoRa with AES-CTR encryption for security. A hardware redesign will integrate sensors on a compact PCB and relocate the device to the ear for better ergonomics and stability.

Software improvements will add features, optimize data processing, and enhance energy efficiency, aiming to deliver a cost-effective, user-beneficial product.

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